Simple Linear Regression

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Model

$$y = \alpha + \beta \cdot x + \varepsilon$$

- y: Dependent variable
- x: Independent variable
- α: y-intercept
- β: Slope
- ε: Unknown, true population errors

Model

$$y = \alpha + \beta \cdot x + \varepsilon$$

- We seek to explain y with x
- "Regress y onto x"
- Observe a sample of y and x
- Estimate α and β
  - Estimates denoted $\hat{\alpha}$ and $\hat{\beta}$
- Do not observe ε
  - Assume $\varepsilon \sim N(0, \sigma^2)$
  - σ^2 unknown and needs to be estimated

Methodology

1. Estimate model
2. Inference: Is the model is good statistically?
   - ANOVA
   - Hypothesis Tests
3. Prediction: Use the model to explain and predict.

Model

Unobserved, true population model

$$y = \alpha + \beta \cdot x + \varepsilon$$

- α, β unobserved true parameters
- ε unobserved population errors
- Observe subsample y and x

Model Estimation

Estimate the model

$$\hat{y} = \hat{\alpha} + \hat{\beta} \cdot x$$

- $\hat{\alpha}, \hat{\beta}$ are estimated
- $\hat{y}$ is the predicted value given x and estimated $\hat{\alpha}$ and $\hat{\beta}$

Sample Error

Sample Error

$$e = y - \hat{y}$$

- e is the sample error from the estimation
- $y = \hat{\alpha} + \hat{\beta} \cdot x + e$

Estimation

- Goal is to choose $\hat{\alpha}, \hat{\beta}$ to find the "best" line
- How do we define best? Minimize the sum of the squared errors

$$\min_{\{\alpha,\beta\}} \sum_{i=1}^{n} e_i^2$$

CAPM analytical solution:

- $\hat{\beta} = \frac{\sigma_{iM}}{\sigma_M^2}$

Regression Application

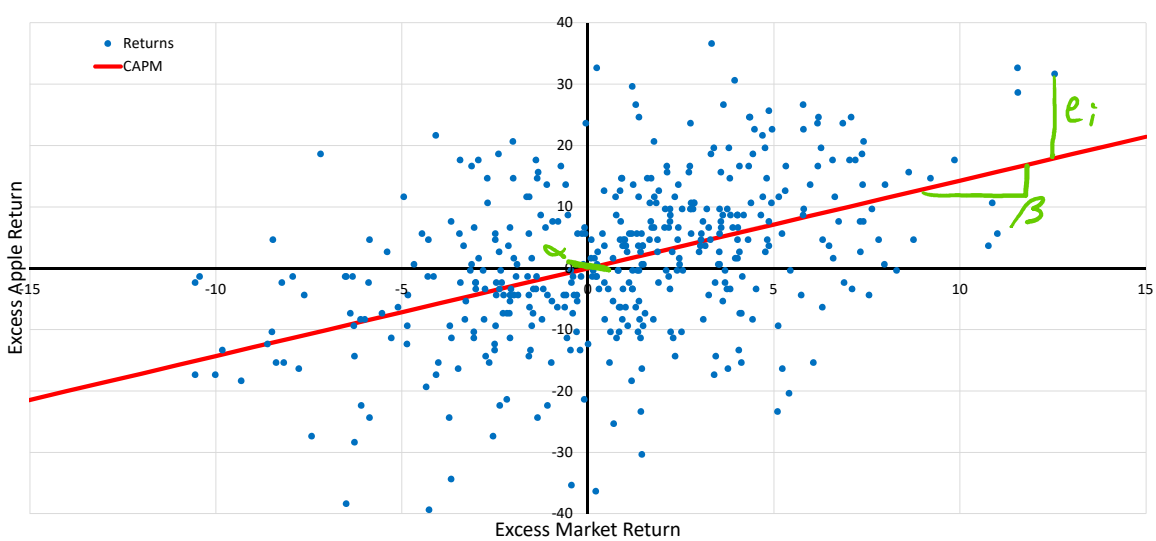
- Mathematically, we have

$$\bar{y} = \hat{\alpha} + \hat{\beta} \cdot \bar{x}$$

- CAPM Interpretation

$$RP_i = \hat{\alpha} + \hat{\beta} \cdot RP_M$$

Apple's Estimated CAPM



Residual Plot: Plot e vs x

Sum of Squares

$$SS = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

- Measure of variation
- $\bar{x}$ is the average of x
- Excel function: $SS = \text{devsq}(x)$

Sum of Squares

SS	Name	Variable	Formula	Interpretation
SSX	Sum of squares of independent variable	x	$\sum_{i=1}^{n}(x_i - \bar{x})^2$	Independent variable
SST	Total sum of squares	y	$\sum_{i=1}^{n}(y_i - \bar{y})^2$	Dependent variable
SSE	Sum of squared errors	e	$\sum_{i=1}^{n}(e_i - \bar{e})^2$	Error
SSR	Sum of squares of regression	$\hat{y}$	$\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$	Regression Model

ANOVA

Decompose variation in data y, i.e. the SST, into the sum of:

- variation explained by the regression model, SSR
- variation unexplained by model, i.e. the error SSE

$$y = \hat{y} + e$$
$$SS(y) = SS(\hat{y}) + SS(e)$$
$$SST = SSR + SSE$$
$$\text{Data} = \text{Model} + \text{Error}$$

ANOVA

	Degrees of Freedom $(\mathbf{df})$	**Sum of Squares** $(\mathbf{SS})$	**Mean Squared** $\left(\mathbf{MS} = \frac{\mathbf{SS}}{\mathbf{df}}\right)$
Regression (Model)	$df_R = 1$	SSR	$MSR = \frac{SSR}{df_R}$
Error	$+df_e = n - 2$	$+SSE$	$MSE = \frac{SSE}{df_e}$
Total (Data)	$= df_T = n - 1$	$= SST$	

Coefficient of Determination

$$r^2 = \frac{SSR}{SST}$$

- Percent variation of y explained by x
- $0 \leq r^2 \leq 1$
- Closer to 1, the better the model

Coefficient of Determination & Correlation

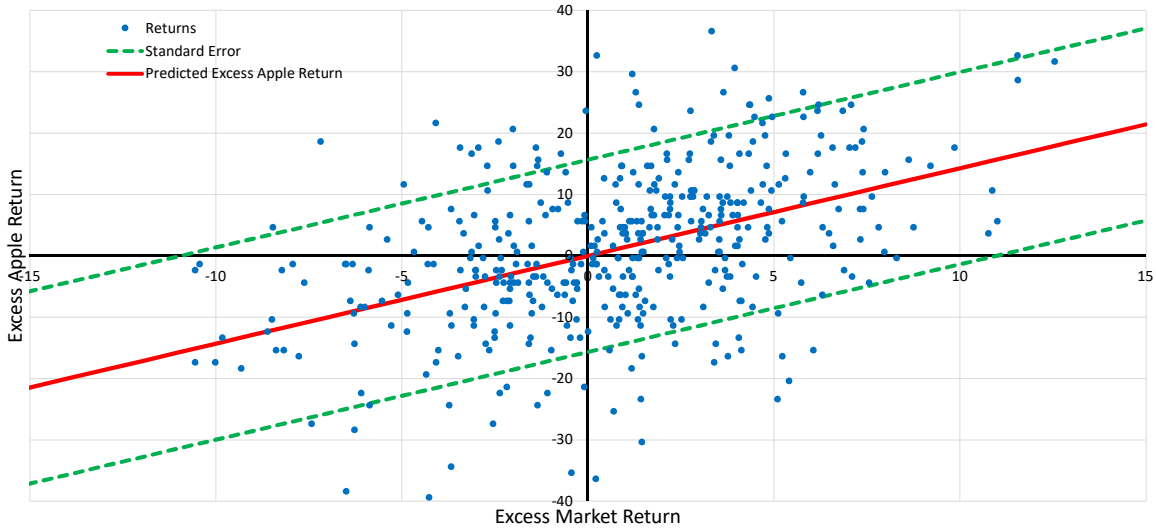
- Correlation $r = \frac{Cov(x,y)}{SD(x) \cdot SD(y)} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$
- Can calculate the coefficient of determination from two perspectives
  - ANOVA: $r^2 = \frac{SSR}{SST}$
  - Correlation: $r^2 = (r)^2$

Standard Error of Regression

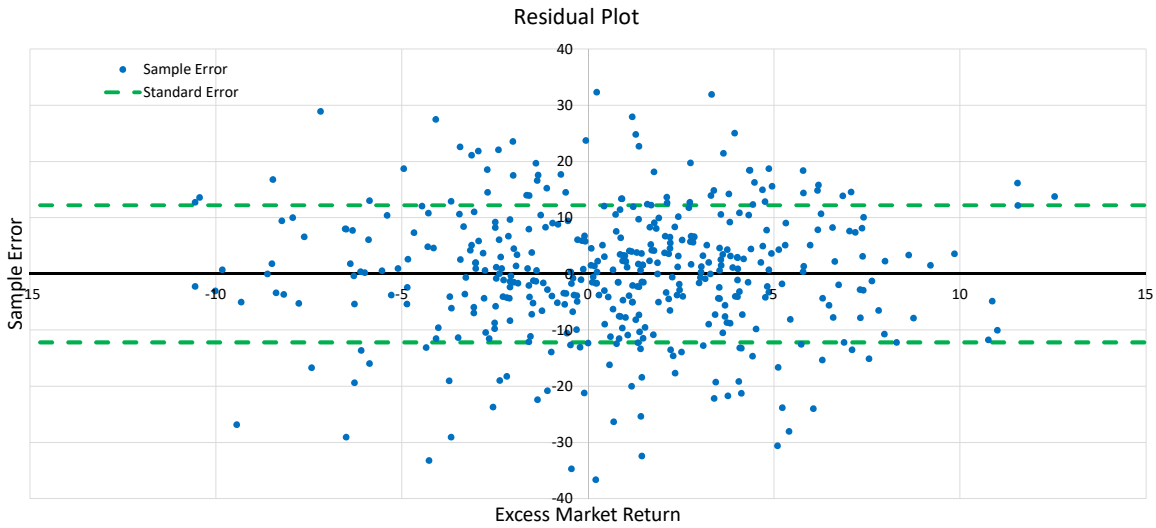
- s is the standard error of the regression
  - The standard error of the regression is approximately the standard deviation of the errors (but we can't use the stdev() formula): $s \approx SD(e)$

$$s = \sqrt{\frac{1}{n-2} \sum_{i=1}^{n} (e_i - \bar{e})^2} = \sqrt{MSE}$$

Standard Error of Regression



Standard Error of Regression



Standard Errors of Estimates

- In practice, we could discuss the standard deviation of
  - The true slope β
    - Unknown, cannot calculate
  - The estimated slope $\hat{\beta}$
    - We will call the standard deviation of an estimate a "standard error" so we do not confuse the two
- Standard error of $\hat{\alpha}$ and $\hat{\beta}$ will allow us to determine how confident we are about our estimations
  - The greater the standard error, the more uncertain we are about the accuracy of our estimate
  - Will be used in inference and hypothesis testing

Standard Errors of Estimates

Standard Error of $\hat{\beta}$

$$SE(\hat{\beta}) = \sqrt{\frac{s^2}{SSX}}$$

- Less error in $\hat{\beta}$ when less noisy data $\downarrow s^2$
- Rely on Excel to calculate $SE(\hat{\alpha})$

Hypothesis Tests

- When we estimate α and β, we want to know how confident we are in our estimates
- Statistical and economically motivated tests
- When we perform a hypothesis test, we will always compare two hypothesis: a null and an alternative
- Goal is to see which hypothesis is supported by data statistically

Hypothesis Tests: t-test

- Null hypothesis:

$$H_0: \beta = \beta_0$$

- Alternative hypothesis:

$$H_a: \beta \neq \beta_0$$

Hypothesis Tests: t-test

- Three methods to conduct hypothesis test
  - All methods will give identical conclusion
  1. Compare x-axis: t-stat vs critical value
  2. Compare y-axis: P-value vs significance level
  3. Examine confidence interval vs null hypothesis

t-distribution

- t-distribution is similar to standard Normal distribution
- Extra parameter: degrees of freedom
  - For hypothesis tests, use df_e
- Excel functions t.dist and t.inv similar to norm.dist and norm.inv
- Probability of $-\infty \leq t \leq a$ is the area under curve from $-\infty$ to a

$$P(-\infty \leq t \leq a) = \mathrm{t.dist}(t, df_e, \mathrm{TRUE})$$

- The value on the x-axis, a, that corresponds to probability $p = P(-\infty \leq t \leq a)$

$$a = \mathrm{t.inv}(p, df_e\ )$$

Calculations from data

- Test statistic
  - x-axis
  - "Standardized deviation of estimate from hypothesized valued"

$$\mathrm{t-stat} = \frac{\hat{\beta} - \beta_0}{SE(\hat{\beta})}$$

- P-value
  - y-axis
  - "Probability of estimating $\hat{\beta}$ given the null is true"

$$\mathrm{P-val} = 2 \times \mathrm{Prob}(t \geq |\mathrm{t-stat}|) = 2 \times \mathrm{t.dist}(-|\mathrm{t-stat}|, df_e)$$

Subjective calculations

- Significance level α
  - y-axis
  - Probability of incorrectly rejecting null
  - Typical levels of 1%, 5%, or 10% in practice
  - If you want to be more confident of result, use smaller α
    - Confidence $= (1-\alpha)100\%$
    - Smaller α is a tougher hurdle to discover a result

- Critical value
  - x-axis

$$\alpha = 2 \times \mathrm{Prob}(t \geq |\mathrm{cv}|) \Rightarrow \mathrm{cv} = \mathrm{t.inv}(1 - \frac{\alpha}{2}, df_e)$$

Type I and II Errors

		True	
		H_0	H_a
Decision	H_0	**Correct** Probability = 1- significance level	**Type II Error** Probability = 1 - power
	H_a	**Type I Error** Probability = Significance Level	**Correct** Probability = Power

Summary

	Data	Subjective
	$\mathrm{P-val} = 2 \times \mathrm{Prob}(t \geq \lvert \mathrm{t-stat} \rvert)$	$\alpha = 2 \times \mathrm{Prob}(t \geq \lvert \mathrm{cv} \rvert)$
x-axis	t-stat $\lvert \mathrm{t-stat} \rvert = \mathrm{t.inv}(1 - \frac{\mathrm{P-val}}{2}, df_e)$	Critical value $\mathrm{cv} = \mathrm{t.inv}(1 - \frac{\alpha}{2}, df_e)$
y-axis	P-value $\mathrm{P-val} = 2 \times \mathrm{t.dist}(-\lvert \mathrm{t-stat} \rvert, df_e)$	Significance level α $\alpha = 2 \times \mathrm{t.dist}\,(-\lvert \mathrm{cv} \rvert, df_e)$

Methods 1 & 2

Method 1: t-stat vs critical value

- Reject null if $|t - stat| \geq |cv|$

Method 2: P-value vs significance level

- Reject null if P-value $<$ significance level

Confidence Intervals

$(1 - \alpha)100\%$ Confidence Interval:

$$\hat{\alpha} \pm \text{cv} \times SE(\hat{\alpha})$$
$$\hat{\beta} \pm \text{cv} \times SE(\hat{\beta})$$

Note we have two alphas:

- Significance level α
  - From statistics
- y-intercept α
  - From finance, called "Jensen's α"
- Different, unrelated values

Method 3: Confidence Interval

- If $\beta_0 \notin CI$ then reject null
  - Same for α_0
- The confidence interval method allows us to quickly conduct numerous hypothesis tests

Prediction

- Predict y with x:

$$\hat{y} = \hat{\alpha} + \hat{\beta} \cdot x$$

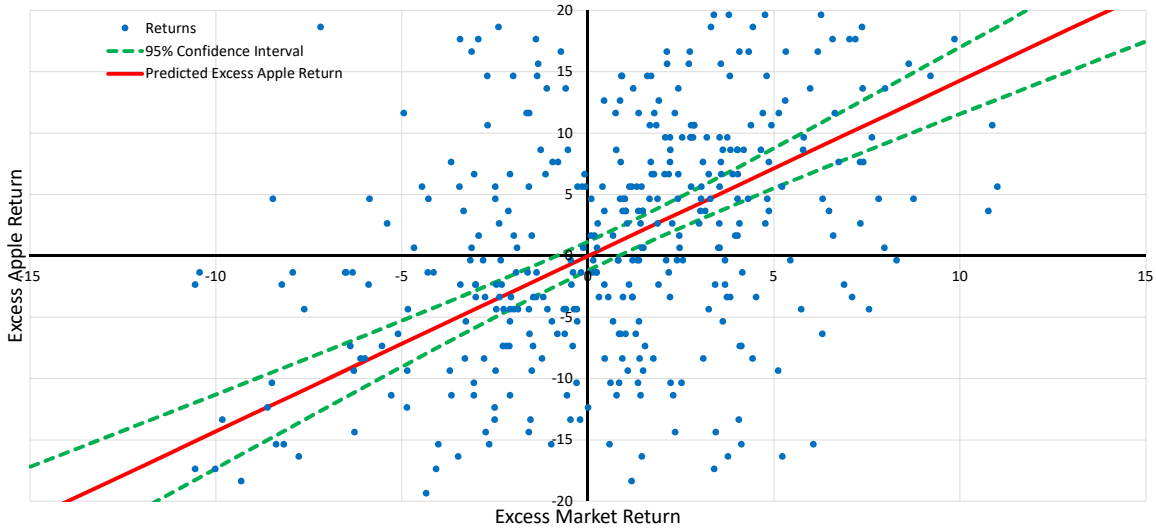
- $(1-\alpha)100\%$ Confidence Interval

$$\hat{y} \pm cv \cdot s \cdot \sqrt{\frac{1}{n} + \frac{(x-\bar{x})^2}{SS_X}}$$

Interpretation

- cv: Increase interval if want to be more confident capturing true value
- s: Increase interval if more model error
- $\frac{1}{n}$: Decrease interval if sample size increases
- $\frac{(x-\bar{x})^2}{SS_X}$: Increase interval as move as from center of data

Apple's Prediction with CI



Inference

Is the CAPM model good statistically?

1. Hypothesis Test: $\beta = 0$
   - $\beta \neq 0 \Rightarrow$ Market risk factor is important
2. ANOVA: r^2
   - $r^2 \ll 100\% \Rightarrow$ Market is not the only important factor

Finance Theory

$\alpha = 0$

- If $\alpha > 0$, two interpretations
  1. Skill
  2. Missing risk factors
  - Empirically we cannot find persistent $\alpha > 0$

$\beta > 1$

- Asset is riskier than market and commands higher risk premium